AISC 5020 Mathematics & Statistics for AI

Lecture 4 | Vector Calculus and Computation Graphs

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Some course logistics

As instructed by the AISCI programme administration, we will need to **keep attendance records starting this class**

- Please sign on the attendance sheet during the break/before you leave
- For AISCI programme record only
- Will NOT count toward your grade or any assessment items in this course
- The attendance for the last three weeks will be considered as **present** by default

(Recall) Some concepts related to $A\mathbf{x} = \mathbf{b}$

- Linear dependence/independence: checking "hidden redundancy" in a set of vectors
- Span: collection of all possible linear combinations

$$\text{span}\{\mathbf{a}_1, \ldots, \mathbf{a}_d\} = \left\{ \sum_{j=1}^{d} x_j \mathbf{a}_j : x_j \in \mathbb{R} \right\}.$$

- Basis: Minimal set of vectors that can span the entire vector space
- Dimension of a vector space: number of vectors in any basis for it
- Rank of a matrix:

$$\text{rank}(A) = \dim(\text{row space of } A) = \dim(\text{column space of } A)$$

- Condition number: measuring how sensitive the solution is to small changes or errors in the input data

$$\kappa_2(A) = \|A\|_2 \left\|A^{-1}\right\|_2 = \frac{\sigma_{\max}}{\sigma_{\min}} \geq 1$$

(Recall) Matrix decomposition methods

Eigen-decomposition:

$$A = VDV^{-1}$$

- Only work for some square matrices (not all)
- D: diagonal matrix with entries $\lambda_1, \ldots, \lambda_n$ being the eigenvalues of A
- V: With column vectors being eigenvectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$

Singular value decomposition (SVD):

$$A = USV^T$$

- Work for general rectangular matrices
- Two versions: Full SVD and Reduced SVD
- S: matrix with diagonal entries storing the singular values $\sigma_1, \sigma_2, \ldots$ of A
- U, V: with orthonormal columns
- Applications: truncated SVD, pseudoinverse, ...

(Recall) Analyzing graph-structured data

Analyzing graph-structured data using linear algebra:

- Graph $G = (V, E)$: V is the set of vertices, and E is the set of undirected/directed edges
- Matrix representation of graphs (Adjacency matrix A, degree matrix D)
- Graph Laplacian $L = D - A$
- Spectrum = all eigenvalues of the graph Laplacian L

$$\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$$

  - We always have $\lambda_1 = 0$
  - The number of zero eigenvalues of L is equal to the number of connected components of G
- λ_2: the algebraic connectivity of the graph
  - > 0 if and only if G is connected
  - More generally, the value of it captures how "well-connected" the graph is

Topics today

Vector Calculus and Computation Graphs

- Derivatives, gradients, Jacobian, Hessians
- Multivariable chain rule
- Taylor series
- Computation graphs
- Automatic differentiation

Why do we need calculus in AI?

Recall that a trained AI model maps inputs and parameters to an output.
We may therefore encounter questions as follows:

- If temperature changes slightly, how does predicted yield change?
- If a coefficient changes, how does the loss change?
- Which direction in parameter space most rapidly increases the loss?

Calculus turns "slightly" and "rapidly" into precise local statements.

Derivatives of single-variable function

Recall that for a single-variable function $y = f(x)$, we can consider the output change divided by the input change (i.e., the slope):

$$\frac{f(x+h) - f(x)}{h}, \qquad h \neq 0,$$

Example: For $f(x) = x^2$, $x = 2$ and $h = 1$, the slope is $(9-4)/1 = 5$.

Taking the input change toward zero, we have the derivative

$$\boxed{f'(x) = \frac{df}{dx} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}}.$$

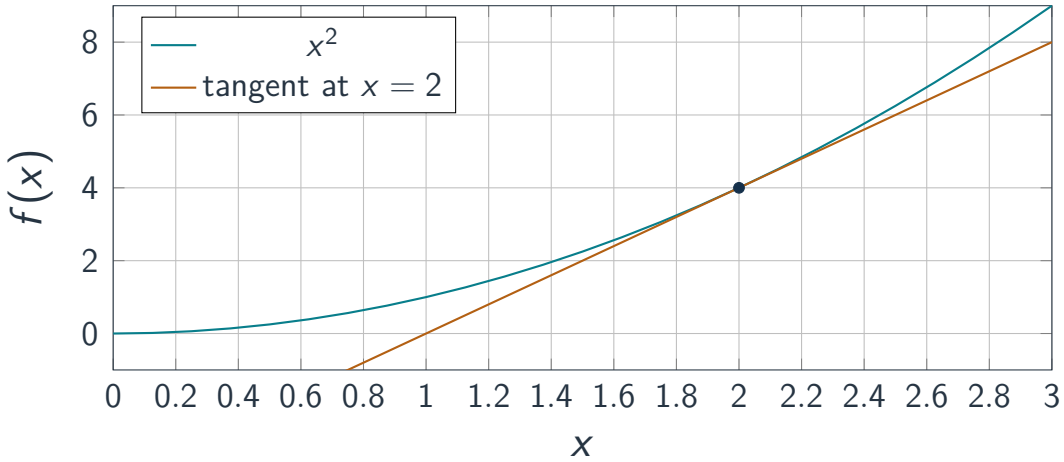
Interpretation of derivatives

Note that the derivatives will give the slope of the tangent line of the function at the chosen point.

Example: For $y = f(x) = x^2$, we have

$$f'(x) = \lim_{h \to 0} \frac{(x+h)^2 - x^2}{h} = \lim_{h \to 0} \frac{2xh + h^2}{h} = \lim_{h \to 0} (2x + h) = 2x.$$

At $x = 2$, $f(2) = 4$ and $f'(2) = 4$, so the tangent line is $y = 4 + 4(x - 2)$. It matches the function's value and slope at the chosen point.



Basic derivative rules

Derivatives of common functions (constant, polynomial, exponential, logarithmic, trigonometric):

$$\frac{d}{dx}c = 0, \qquad \frac{d}{dx}x^p = px^{p-1}, \qquad \frac{d}{dx}e^x = e^x, \qquad \frac{d}{dx}\ln x = \frac{1}{x}.$$

$$\frac{d}{dx}\sin x = \cos x, \qquad \frac{d}{dx}\cos x = -\sin x, \qquad \frac{d}{dx}\tan x = -\sec^2 x.$$

Basic rules of operation:

$$\boxed{\frac{d}{dx}\Big[af(x) + bg(x)\Big] = af'(x) + bg'(x)}$$

for constants a, b.

Example:

$$\frac{d}{dx}(3x^2 - 2\sin x + 5e^x + 3) = 3(2x) - 2\cos x + 5e^x + 0 = 6x - 2\cos x + 5e^x.$$

Basic derivative rules

Product rule:

$$\boxed{(fg)' = f'g + fg'}.$$

Example: For $h(x) = x^2 e^x$,

$$h'(x) = 2xe^x + x^2 e^x.$$

Chain rule: If $u = f(x)$ and $y = g(u)$, then

$$\boxed{\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx} = g'(f(x))f'(x)}.$$

Example: For $y = (3x+1)^2$, set $u = 3x + 1$.

$$\frac{dy}{du} = 2u, \qquad \frac{du}{dx} = 3, \qquad \frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}(2u)(3) = 6(3x+1).$$

Example

Differentiate the following functions with respect to x:

$$f(x) = 2x^3 - 4x + 1, \qquad g(x) = \ln(1 + x^2), \qquad h(x) = xe^{2x}.$$

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$$f(x) = 2x^3 - 4x + 1, \qquad g(x) = \ln(1 + x^2), \qquad h(x) = xe^{2x}.$$

Answer: For f, we can use the basic rules of operation:

$$f'(x) = 2(3x^2) - 4 + 0 = 6x^2 - 4$$

For g, we can use the chain rule (considering $u = 1 + x^2$ and $g(u) = \ln(u)$):

$$g'(x) = \frac{dg}{du}\frac{du}{dx} = \frac{1}{u}(2x) = \frac{2x}{1 + x^2}.$$

For h, we can use both the product rule and chain rule (considering $u = 2x$ and so $\frac{d}{dx}e^{2x} = \frac{d(e^u)}{du}\frac{du}{dx}$):

$$h'(x) = x'e^{2x} + x(e^{2x})' = (1)e^{2x} + x(2e^{2x}) = e^{2x} + 2xe^{2x}$$

Taylor series

Derivatives can also be used for representing or approximating a function.

Specifically, the **Taylor series** (also known as the **Taylor expansion**) of a function f is an infinite sum of terms that are expressed in terms of the function's derivatives:

$$f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

When $a = 0$, the series is also called a **Maclaurin series** and takes the form:

$$f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \cdots = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n$$

Taylor series

Examples:

$$
\begin{aligned}
e^x &= \sum_{n=0}^{\infty} \frac{x^n}{n!} &&= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots && \text{for all } x \\
\sin x &= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} &&= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots && \text{for all } x \\
\cos x &= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} &&= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots && \text{for all } x \\
\frac{1}{1-x} &= \sum_{n=0}^{\infty} x^n &&= 1 + x + x^2 + x^3 + \cdots && \text{for all } |x| < 1 \\
\ln(x) &= \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} (x-1)^n &&= (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \cdots && \text{for all } |x-1| < 1
\end{aligned}
$$

Taylor polynomial

The partial sum formed by the first $n+1$ terms of a Taylor series is called the ***n*-th order Taylor polynomial** of the function:

$$T_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n$$

Taylor polynomials can be used for approximating a function.

For example, the first-order Taylor polynomial of f around $x = a$ gives a linear approximation:

$$f(x) \approx f(a) + f'(a)(x-a)$$

Example: For $f(x) = x^2$ around $x = 2$,

$$f(x) \approx f(2) + f'(2)(x-2) = 4 + 4(x-2).$$

With $x = 2.1$, the approximation is 4.4, while the exact value is $2.1^2 = 4.41$.

Taylor polynomial

The second-order Taylor polynomial of f around $x = a$ gives a quadratic approximation:

$$f(x) \approx f(a) + f'(a)(x - a) + \frac{f''(a)}{2}(x - a)^2$$

Example: For $f(x) = \cos(x)$ around $x = 0$, we have

$$f(0) = \cos(0) = 1,$$

$$f'(x) = -\sin(x) \Rightarrow f'(0) = -\sin(0) = 0,$$

$$f''(x) = -\cos(x) \Rightarrow f''(0) = -1,$$

and hence

$$f(x) \approx f(0) + f'(0)(x - 0) + \frac{f''(0)}{2}(x - 0)^2 = 1 - \frac{x^2}{2}.$$

Example

Let $f(x) = e^x \sin x$.

- Estimate $f(0.01)$ using the linear and quadratic approximations.
- Compare the results with the exact value.

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- Estimate $f(0.01)$ using the linear and quadratic approximations.
- Compare the results with the exact value.

Answer: Differentiating gives $f'(x) = e^x(\sin x + \cos x)$ and $f''(x) = 2e^x \cos x$.
Evaluating at the expansion point yields

$$f(0) = 0, \qquad f'(0) = 1, \qquad f''(0) = 2.$$

Hence the linear approximation is

$$f(0.01) \approx f(0) + f'(0)(0.01 - 0) = 0.01$$

and the quadratic approximation is

$$f(0.01) \approx f(0) + f'(0)(0.01 - 0) + \frac{f''(0)}{2}(0.01 - 0)^2 = 0.0101.$$

The exact value is $f(0.01) = e^{0.01} \sin(0.01) = 0.010100333\ldots$, so the quadratic approximation already captures the first six significant digits.

Multivariable functions

Recall that a single-variable function $y = f(x)$ takes one single input x and outputs a value y.
A general **multivariable function** is a map

$$f : U \subseteq \mathbb{R}^n \to \mathbb{R}^m,$$

where the set U is the domain. In other words, it can take a vector of n values and output a vector of m values.

- If $m = 1$, we call f a scalar-valued function.
- If $m \geq 2$, we call $\mathbf{f} = (f_1, \ldots, f_m)$ a vector-valued function.

Examples:

- A function $f : \mathbb{R}^2 \to \mathbb{R}$:
$$f(x, y) = x^2 + 3xy + y^2.$$
- A function $T : \mathbb{R}^3 \to \mathbb{R}$:
$$T(x, y, z) = 20 + 3x - y^2 + z$$
- Polar change of coordinates (vector-valued) function $f : [0, \infty) \times [0, 2\pi) \to \mathbb{R}^2$:
$$f(r, \theta) = (r\cos\theta,\ r\sin\theta).$$

Partial derivatives

We can extend the concept of derivatives in single-variable calculus to partial derivatives for multivariable functions!

The idea is to differentiate with respect to one of the variables and keeping all other variables fixed and treating them as constant.

Formally, the **partial derivative** of f with respect to the i-th coordinate at $\mathbf{a}$ is

$$\boxed{\frac{\partial f}{\partial x_i}(\mathbf{a}) = \lim_{h \to 0} \frac{f(\mathbf{a} + h\mathbf{e}_i) - f(\mathbf{a})}{h}}$$

In particular, for $f = f(x, y)$, we have

$$\frac{\partial f}{\partial x}(x, y) = f_x(x, y) = \lim_{h \to 0} \frac{f(x + h, y) - f(x, y)}{h}$$

$$\frac{\partial f}{\partial y}(x, y) = f_y(x, y) = \lim_{h \to 0} \frac{f(x, y + h) - f(x, y)}{h}$$

Partial derivatives

Example: For $f(x, y) = e^x \sin y$, we have

$$f_x = e^x \sin y$$

$$f_y = e^x \cos y.$$

Example: For $g(x, y) = x^2 y + \sin(xy)$,

$$g_x = 2xy + y\cos(xy)$$

$$g_y = x^2 + x\cos(xy).$$

Partial derivatives

Example: For $f(x, y) = x^2 + 3xy + y^2$,

$$\frac{\partial f}{\partial x} = 2x + 3y + 0 = 2x + 3y$$

$$\frac{\partial f}{\partial y} = 0 + 3x + 2y = 3x + 2y.$$

At $(x, y) = (1, 2)$,

$$f_x = 2(1) + 3(2) = 8$$

$$f_y = 3(1) + 2(2) = 7.$$

This shows that a small change Δx alone gives approximately $8\Delta x$ change in f, while a small change Δy alone gives approximately $7\Delta y$.
These values describe local sensitivity at this particular point.

Gradients

For a scalar function $f : \mathbb{R}^n \to \mathbb{R}$, the **gradient** of f is given by

$$\nabla f(\mathbf{x}) = \left(\frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_n}\right) \in \mathbb{R}^n.$$

Sometimes we may also write it in the column vector form:

$$\nabla f(\mathbf{x}) = \begin{bmatrix} \partial f/\partial x_1 \\ \vdots \\ \partial f/\partial x_n \end{bmatrix} \in \mathbb{R}^n.$$

Example: For $f(x, y) = x^2 + 3xy + y^2$, we have

$$\nabla f = (f_x, f_y) = (2x + 3y, 3x + 2y)$$

and

$$\nabla f(1, 2) = (8, 7)$$

Gradient and first-order (linear) approximation

Analogous to the single-variable case, for a differentiable multivariable function f, we have the **first-order (linear) approximation** given by

$$\boxed{f(\mathbf{x} + \Delta\mathbf{x}) \approx f(\mathbf{x}) + \nabla f(\mathbf{x})^{\mathsf{T}} \Delta\mathbf{x}}$$

Example:
For $f(x, y) = x^2 + 3xy + y^2$, recall that

$$\nabla f(1, 2) = (8, 7).$$

Therefore, if $\Delta\mathbf{x} = (0.01, -0.02)^{\mathsf{T}}$, then

$$\Delta f = f((1, 2) + \Delta\mathbf{x}) - f(1, 2) \approx 8(0.01) + 7(-0.02) = -0.06.$$

Directional derivatives

Note that the partial derivatives $\frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_n}$ are all with respect to the direction of the standard coordinate vectors.

More generally, we can consider the derivatives along other chosen directions using directional derivatives.

Formally, the **directional derivative** of f in a direction $\mathbf{v}$ is

$$\boxed{D_{\mathbf{v}} f(\mathbf{x}) = \frac{d}{dt} f(\mathbf{x} + t\mathbf{v})\Big|_{t=0} = \nabla f(\mathbf{x})^{\mathsf{T}} \mathbf{v}.}$$

If $\|\mathbf{v}\|_2 = 1$, it measures change per unit distance along that direction.

Without unit normalization, it measures change per unit path parameter t.

Example

For $f(x, y) = x^2 y$, find the directional derivative of f in the direction of $\mathbf{v} = (\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}})$ at the point $(3, 2)$.

Example

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Answer: We first compute:

$$f_x = 2xy \Rightarrow f_x(3,2) = 12,$$
$$f_y(x,y) = x^2 \Rightarrow f_y(3,2) = 9.$$

Therefore, the directional derivative is

$$D_{\mathbf{v}} f(3,2) = (12,9) \cdot (\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}) = \frac{12}{\sqrt{2}} - \frac{9}{\sqrt{2}} = \frac{3}{\sqrt{2}}$$

Gradient and steepest ascent/descent

Recall that

$$f(\mathbf{x}+\mathbf{v}) \approx f(\mathbf{x}) + \nabla f(\mathbf{x})^{\mathsf{T}}\mathbf{v}.$$

For unit $\mathbf{v}$, by the Cauchy–Schwarz inequality ($|\mathbf{a}\cdot\mathbf{b}| \le \|\mathbf{a}\|_2\|\mathbf{b}\|_2$), we have

$$|\nabla f^{\mathsf{T}}\mathbf{v}| \le \|\nabla f\|_2 \, \|\mathbf{v}\|_2 = \|\nabla f\|_2 .$$

If $\nabla f \neq 0$, equality occurs at

$$\boxed{\mathbf{v} = \pm\frac{\nabla f}{\|\nabla f\|_2}}.$$

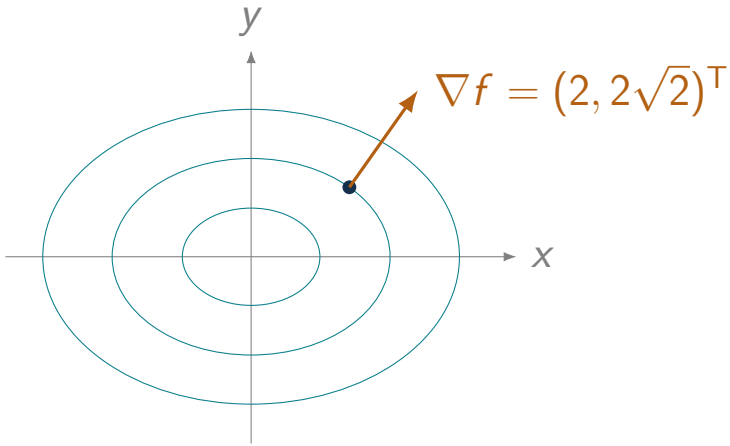
(Explanation: Note that the equality in the Cauchy–Schwarz inequality $|\mathbf{a}\cdot\mathbf{b}| \le \|\mathbf{a}\|_2\|\mathbf{b}\|_2$ holds if and only if $\mathbf{a}$ and $\mathbf{b}$ are linearly dependent, so here we can write $\mathbf{v} = t\nabla f$ for some scalar t. Then since $\mathbf{v}$ is a unit vector, the only possibilities are $\mathbf{v} = \pm\frac{\nabla f}{\|\nabla f\|_2}$.)

This shows that

- ∇f points in the direction of steepest ascent (with $\nabla f(\mathbf{x})^{\mathsf{T}}\mathbf{v} = \frac{\nabla f(\mathbf{x})^{\mathsf{T}}\nabla f}{\|\nabla f\|_2} = \|\nabla f\|_2$), and
- $-\nabla f$ points in the direction of steepest descent (with $\nabla f(\mathbf{x})^{\mathsf{T}}\mathbf{v} = \frac{\nabla f(\mathbf{x})^{\mathsf{T}}(-\nabla f)}{\|\nabla f\|_2} = -\|\nabla f\|_2$).

Gradient and steepest ascent/descent

Geometrically, we can see that gradients are perpendicular to level curves of a function.



Contours of $f(x, y) = x^2 + 2y^2$

Along a differentiable curve on which f stays constant, the directional derivative along its tangent is zero. Therefore the gradient is perpendicular to that tangent when the gradient is nonzero.

Gradient and critical points

Recall that in single-variable calculus, a function f has a critical point at $x = a$ if $f'(a) = 0$ or f' is undefined at a.

Analogously, for a multivariable function $f : \mathbb{R}^n \to \mathbb{R}$, $\mathbf{x} = \mathbf{a}$ is said to be a **critical point** if the gradient $\nabla f(\mathbf{a}) = \mathbf{0}$ or is undefined (i.e., some partial derivatives are undefined).

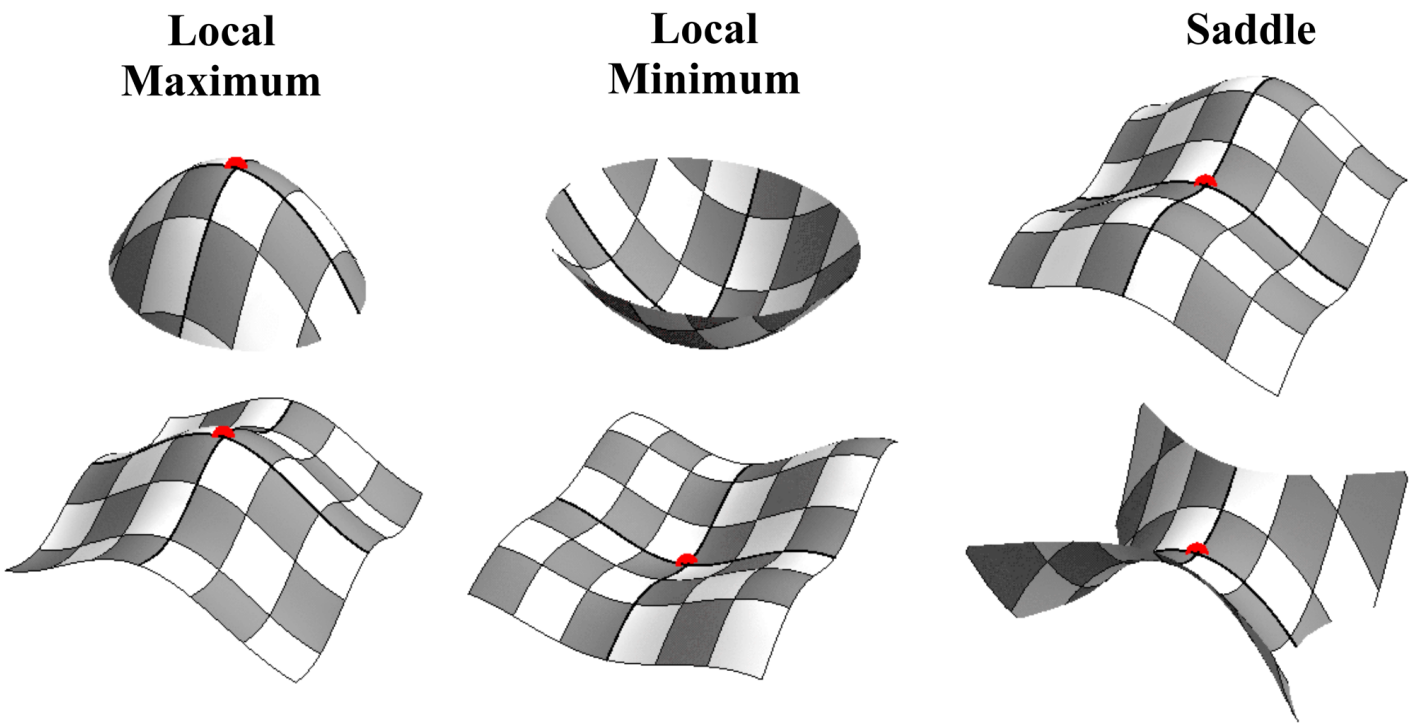
The critical values are the values of the function at the critical points.

A critical point (where the function is differentiable) may be either

- a local maximum,
- a local minimum, or
- a saddle point.

The different cases may be distinguished by considering the Hessian matrix, which will be covered later in this lecture.

Gradient and critical points



(Source: Wikipedia)

Example

Example: For $f(x, y) = x^2 + y^2$ we have

$$\nabla f = (2x, 2y)$$

At $(1, 1)$, the gradient is $(2, 2)$, pointing radially outward, perpendicular to the circle $x^2 + y^2 = 2$. The steepest-ascent slope is

$$(2, 2) \cdot \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) = 2\sqrt{2}$$

Example: For $f(x, y) = e^x \sin y$, we have

$$\nabla f = (e^x \sin y,\ e^x \cos y), \qquad \nabla f(0, 0) = (0, 1).$$

Near the origin, the function increases most rapidly in the positive y-direction.

Example

Let $f(x, y) = x^2 + 2y^2$ at $(1, 1)$.

- Find the gradient of f at $(1, 1)$.
- Find the directional derivative along $\mathbf{v} = (1, -1)^\mathsf{T}/\sqrt{2}$.
- Give a unit direction of steepest descent.
- Approximate the change for $\Delta\mathbf{x} = (0.1, -0.1)^\mathsf{T}$ near the point $(1, 1)$.

Example

Let $f(x, y) = x^2 + 2y^2$ at $(1, 1)$.

- Find the gradient of f at $(1, 1)$.
- Find the directional derivative along $\mathbf{v} = (1, -1)^\mathsf{T}/\sqrt{2}$.
- Give a unit direction of steepest descent.
- Approximate the change for $\Delta\mathbf{x} = (0.1, -0.1)^\mathsf{T}$ near the point $(1, 1)$.

Answer: We have

$$\nabla f = (2, 4)^\mathsf{T}, \qquad D_\mathbf{v} f = (2, 4) \cdot (\frac{-1}{\sqrt{2}}, -\frac{-1}{\sqrt{2}}) = \frac{2-4}{\sqrt{2}} = -\sqrt{2}.$$

The steepest-descent unit direction is $-\dfrac{\nabla f}{\|\nabla f\|_2} = -(\dfrac{2}{\sqrt{2^2+4^2}}, \dfrac{4}{\sqrt{2^2+4^2}}) = (\dfrac{-1}{\sqrt{5}}, \dfrac{-2}{\sqrt{5}})$.

For $\Delta\mathbf{x} = (0.1, -0.1)^\mathsf{T}$, the approximated change is

$$\Delta f = f(\mathbf{x} + \Delta\mathbf{x}) - f(\mathbf{x}) \approx \nabla f^\mathsf{T} \Delta\mathbf{x} = 2(0.1) + 4(-0.1) = -0.2.$$

The exact change is $1.21 + 2(0.81) - 3 = -0.17$.

Jacobian matrix for vector-valued functions

Note that the gradient notation only works for scalar-valued functions $f : \mathbb{R}^n \to \mathbb{R}$. What if we have a vector-valued function $f : \mathbb{R}^n \to \mathbb{R}^m$?
In this case, we need a matrix instead of a vector to store all partial derivatives!
Formally, let $f : U \subseteq \mathbb{R}^n \to \mathbb{R}^m$ with

$$f = (f_1, \ldots, f_m).$$

The **Jacobian** matrix of f at $\mathbf{a}$ is an $m \times n$ matrix with the (i, j) entry being $\dfrac{\partial f_i}{\partial x_j}$. In other words, we have

$$J_f(\mathbf{x}) = \begin{pmatrix} \dfrac{\partial f_1}{\partial x_1} & \cdots & \dfrac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \dfrac{\partial f_m}{\partial x_1} & \cdots & \dfrac{\partial f_m}{\partial x_n} \end{pmatrix}$$

If $m = n$, J_f is a square matrix, and we can also consider the Jacobian determinant $\det J_f$.

Jacobian matrix for vector-valued functions

Example: Let

$$f(x,y) = \begin{pmatrix} f_1(x,y) \\ f_2(x,y) \end{pmatrix} = \begin{pmatrix} x^2 + y \\ xy \end{pmatrix}.$$

The Jacobian matrix of f is

$$J_f(x,y) = \begin{pmatrix} \dfrac{\partial f_1}{\partial x} & \dfrac{\partial f_1}{\partial y} \\ \dfrac{\partial f_2}{\partial x} & \dfrac{\partial f_2}{\partial y} \end{pmatrix} = \begin{pmatrix} 2x & 1 \\ y & x \end{pmatrix}.$$

The Jacobian determinant is

$$\det J_f = (2x)(x) - y(1) = 2x^2 - y$$

Jacobian matrix for vector-valued functions

Example: For the polar change of coordinates function $f : [0, \infty) \times [0, 2\pi) \to \mathbb{R}^2$ that transforms polar coordinates (r, φ) to Cartesian coordinates (x, y), we have

$$(x, y) = f(r, \varphi) = (r \cos \varphi, r \sin \varphi)$$

The Jacobian matrix of f is

$$J_f(r, \varphi) = \begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \varphi} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \varphi} \end{pmatrix} = \begin{pmatrix} \cos \varphi & -r \sin \varphi \\ \sin \varphi & r \cos \varphi \end{pmatrix}$$

The Jacobian determinant of f is

$$\det J_f = (\cos \varphi)(r \cos \varphi) - (\sin \varphi)(-r \sin \varphi) = r(\cos^2 \varphi + \sin^2 \varphi) = r.$$

Jacobian matrix for vector-valued functions

Example: Consider a function $f : \mathbb{R}^2 \to \mathbb{R}^3$, with $(x, y) \mapsto (f_1(x, y), f_2(x, y), f_3(x, y))$, given by

$$f(x, y) = \begin{pmatrix} f_1(x, y) \\ f_2(x, y) \\ f_3(x, y) \end{pmatrix} = \begin{pmatrix} x^2 y \\ 4x + \sin y \\ 3y \end{pmatrix}.$$

The Jacobian matrix of f is

$$J_f(x, y) = \begin{pmatrix} \dfrac{\partial f_1}{\partial x} & \dfrac{\partial f_1}{\partial y} \\ \dfrac{\partial f_2}{\partial x} & \dfrac{\partial f_2}{\partial y} \\ \dfrac{\partial f_3}{\partial x} & \dfrac{\partial f_3}{\partial y} \end{pmatrix} = \begin{pmatrix} 2xy & x^2 \\ 4 & \cos y \\ 0 & 3 \end{pmatrix}$$

Jacobian and first-order (linear) approximation

For a scalar-valued multivariable function $f : \mathbb{R}^n \to \mathbb{R}$, we can use the gradient for expressing its linear approximation.
Analogously, for a vector-valued function $f : \mathbb{R}^n \to \mathbb{R}^m$, the Jacobian J_f gives the following **first-order (linear) approximation** of f near a point $\mathbf{a}$:

$$\boxed{f(\mathbf{a}+\mathbf{h}) \approx f(\mathbf{a}) + J_f(\mathbf{a})\,\mathbf{h}}$$

Example: Let $f(x,y) = \begin{pmatrix} f_1(x,y) \\ f_2(x,y) \end{pmatrix} = \begin{pmatrix} x^2+y \\ xy \end{pmatrix}$. At $(1,2)$, we have $J_f = \begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix}$. Therefore, for $\Delta\mathbf{x} = (0.1, -0.1)^\mathsf{T}$, the approximated error is

$$\Delta f = f((1,2)+\Delta\mathbf{x}) - f(1,2) \approx J_f \Delta\mathbf{x} = \begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 0.1 \\ -0.1 \end{pmatrix} = (0.1, 0.1)^\mathsf{T}.$$

We can also check that the actual change is $(0.11, 0.09)^\mathsf{T}$, so the first-order approximation already gives a small error.

Jacobian and first-order (linear) approximation

Example:
For an affine map

$$f(\mathbf{x}) = W\mathbf{x} + \mathbf{b}$$

where W is a constant matrix and $\mathbf{b}$ is a constant vector, we have

$$J_f(x) = W.$$

In this case, the first-order approximation is

$$f(\mathbf{0}) + J_f(\mathbf{0})\mathbf{h} = \mathbf{b} + W\mathbf{h} = f(\mathbf{h}),$$

which recovers f exactly.

Jacobian and first-order (linear) approximation

Example:

If $f(x_1, \ldots, x_n) = \begin{pmatrix} f_1(x_1) \\ f_2(x_2) \\ \vdots \\ f_n(x_n) \end{pmatrix}$, i.e., each function only depends on a different coordinate, then the Jacobian J_f is a diagonal matrix with diagonal entries $\frac{\partial f_1}{\partial x_1}, \frac{\partial f_2}{\partial x_2}, \ldots, \frac{\partial f_n}{\partial x_n}$. The off-diagonal entries are zero because f_i does not depend on x_j for $i \neq j$. We have

$$f(\mathbf{x} + \Delta\mathbf{x}) \approx f(\mathbf{x}) + J_f(\mathbf{x})\Delta\mathbf{x} = \begin{pmatrix} f_1(x_1) \\ f_2(x_2) \\ \vdots \\ f_n(x_n) \end{pmatrix} + \begin{pmatrix} \frac{\partial f_1}{\partial x_1}\Delta x_1 \\ \frac{\partial f_2}{\partial x_2}\Delta x_2 \\ \vdots \\ \frac{\partial f_n}{\partial x_n}\Delta x_n \end{pmatrix} = \begin{pmatrix} f_1(x_1) + \frac{\partial f_1}{\partial x_1}\Delta x_1 \\ f_2(x_2) + \frac{\partial f_2}{\partial x_2}\Delta x_2 \\ \vdots \\ f_n(x_n) + \frac{\partial f_n}{\partial x_n}\Delta x_n \end{pmatrix}$$

Hessian matrix

The **Hessian matrix** of a function $f : \mathbb{R}^n \to \mathbb{R}$ is the $n \times n$ matrix of second partial derivatives:

$$H_f(\mathbf{x}) = D^2 f(\mathbf{x}) = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1 \partial x_1} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n \partial x_n} \end{pmatrix}$$

In other words, for all i, j, we have

$$H_f(x)_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial}{\partial x_i}\left(\frac{\partial f}{\partial x_j}\right).$$

If f is C^2 (meaning its second-order partial derivatives are continuous), then the Hessian matrix is always symmetric (i.e., $\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}$).

Hessian matrix

Example: For $f = x^2 + 3xy + y^2$, we have

$$\nabla f = \begin{pmatrix} f_x \\ f_y \end{pmatrix} = \begin{pmatrix} 2x + 3y \\ 3x + 2y \end{pmatrix}, \qquad H_f = \begin{pmatrix} f_{xx} & f_{yx} \\ f_{xy} & f_{yy} \end{pmatrix} = \begin{pmatrix} 2 & 3 \\ 3 & 2 \end{pmatrix}.$$

It can be observed that H_f is symmetric.

Example: For $f(x, y) = e^x \sin y$, we have

$$f_x = e^x \sin y, \qquad f_y = e^x \cos y,$$

$$f_{xx} = e^x \sin y, \qquad f_{xy} = f_{yx} = e^x \cos y, \qquad f_{yy} = -e^x \sin y.$$

Therefore,

$$H_f(x, y) = \begin{pmatrix} e^x \sin y & e^x \cos y \\ e^x \cos y & -e^x \sin y \end{pmatrix}$$

Hessian matrix and second-order approximation

Note that the Hessian is the derivative of the gradient: it records how the slope itself changes. The associated quadratic form

$$\Delta \mathbf{x}^\top H_f(\mathbf{x})\, \Delta \mathbf{x}$$

is the second-order correction to the linear approximation of f. More specifically, we have the following **second-order (quadratic) approximation** of f:

$$\boxed{f(\mathbf{x} + \Delta \mathbf{x}) \approx f(\mathbf{x}) + \nabla f(\mathbf{x})^\mathsf{T} \Delta \mathbf{x} + \frac{1}{2} \Delta \mathbf{x}^\mathsf{T} H_f(\mathbf{x})\, \Delta \mathbf{x}}$$

Moreover, we can further use the Hessian to analyze whether a critical point is a local maximum, local minimum, or saddle.

Hessian and the second partial derivative test

For a differentiable function $f(x, y)$ of **two variables**, we can consider the value of the determinant of the Hessian matrix H_f:

$$D(x, y) = \det(H_f(x, y)) = f_{xx}(x, y)f_{yy}(x, y) - (f_{xy}(x, y))^2$$

Suppose that (a, b) is a critical point of f (with $f_x(a, b) = f_y(a, b) = 0$). Then the **second partial derivative test** asserts the following:

- If $D(a, b) > 0$ and $f_{xx}(a, b) > 0$ then (a, b) is a local minimum of f.
- If $D(a, b) > 0$ and $f_{xx}(a, b) < 0$, then (a, b) is a local maximum of f.
- If $D(a, b) < 0$, then (a, b) is a saddle point of f.
- If $D(a, b) = 0$, then the point (a, b) could be any of a minimum, maximum, or saddle point (that is, the test is inconclusive).

Note: This approach only works for functions of two variables. For general multivariable functions, we need to consider the eigenvalues of the Hessian matrix (see next page).

Hessian and the second partial derivative test

For general multivariable functions $f : \mathbb{R}^n \to R$, let $\mathbf{a}$ be a critical point for which the Hessian matrix $H_f(\mathbf{a})$ is invertible. The **second partial derivative test** is as follows:

- If the Hessian is positive definite (equivalently, has all eigenvalues positive) at $\mathbf{a}$, then f attains a local minimum at $\mathbf{a}$.
- If the Hessian is negative definite (equivalently, has all eigenvalues negative) at $\mathbf{a}$, then f attains a local maximum at $\mathbf{a}$.
- If the Hessian has both positive and negative eigenvalues, then $\mathbf{a}$ is a saddle point for f.
- In those cases not listed above, the test is inconclusive.

Hessian and the second partial derivative test

Example: For $f = x^2 + 3xy + y^2$,

$$\nabla f = 0 \Rightarrow (2x + 3y, 3x + 2y) = 0 \Rightarrow (x, y) = (0, 0).$$

It is easy to see that:

- Along the direction $y = x$, $f = 5x^2 \geq 0$.
- Along the direction $y = -x$, $f = -x^2 \leq 0$.

This shows that there are both increasing and decreasing directions at the origin, i.e., it is a saddle.

We can verify this by applying the second partial derivative test. Note that the Hessian is:

$$H_f = \begin{pmatrix} 2 & 3 \\ 3 & 2 \end{pmatrix}.$$

Therefore, we have:

- (Determinant approach) $D = \det H_f = 2^2 - 3^2 = -5 < 0$, and hence it is a saddle point.
- (Eigenvalue approach) The eigenvalues of H_f are $\lambda = 5$ and $\lambda = -1$. The mixed signs indicate that the point is a saddle point.

Hessian and the second partial derivative test

Example: For $f(x, y) = x^3 + y^3 - 3xy$, we have

$$\nabla f = (f_x, f_y) = (3x^2 - 3y, 3y^2 - 3x)$$

To find the critical points, we have

$$(3x^2 - 3y, 3y^2 - 3x) = (0, 0) \Rightarrow \begin{cases} x^2 - y = 0 \\ y^2 - x = 0 \end{cases}$$

This gives

$$(x^2)^2 - x = 0 \Rightarrow x(x^3 - 1) = 0 \Rightarrow x = 0, 1,$$

and hence we have two critical points $(0, 0)$ and $(1, 1)$.

Hessian and the second partial derivative test

(Continued) We further get the second derivatives:

$$f_{xx} = \frac{\partial}{\partial x}(3x^2 - 3y) = 6x, \qquad f_{yy} = \frac{\partial}{\partial y}(3y^2 - 3x) = 6y, \qquad f_{xy} = \frac{\partial}{\partial y}(3x^2 - 3y) = -3$$

At point $(0, 0)$, we have

$$H_f = \begin{pmatrix} 0 & -3 \\ -3 & 0 \end{pmatrix}$$

and $D = \det H_f = 0 - 9 < 0$, which shows that $(0, 0)$ is a saddle point.

At point $(1, 1)$, we have

$$H_f = \begin{pmatrix} 6 & -3 \\ -3 & 6 \end{pmatrix}$$

and $D = \det H_f = 36 - 9 = 27 > 0$. Also, $f_{xx}(1, 1) = 6 > 0$. Hence, $(1, 1)$ is a local minimum.

Hessian and the second partial derivative test

We remark that there can be situations where the second partial derivative test is inconclusive.

Example: For $f(x, y) = x^4 + y^4$, we have

$$f_x = 4x^3, \qquad f_y = 4y^3$$

and hence a critical point is $(0, 0)$.
The Hessian matrix is

$$H_f = \begin{pmatrix} 12x^2 & 0 \\ 0 & 12y^2 \end{pmatrix}$$

and hence $H_f(0,0) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$, which gives repeated eigenvalues $\lambda_1 = \lambda_2 = 0$. We can check that it does not satisfy any of the first three cases in the second partial derivative test, and so the test is inconclusive.

Example

For each function, find the critical points and classify their types:

$$f(x,y) = (x-1)^2 + 2(y+2)^2, \qquad g(x,y) = x^2 - y^2.$$

Example

For each function, find the critical points and classify their types:

$$f(x,y) = (x-1)^2 + 2(y+2)^2, \qquad g(x,y) = x^2 - y^2.$$

Answer:

For f, we have $\nabla f = (2(x-1), 4(y+2))^{\mathsf{T}}$, so the only critical point is $(x,y) = (1,-2)$.

Its Hessian is $H_f = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}$, with eigenvalues $\lambda = 2, 4$. Since both eigenvalues are positive, we conclude that $(1,-2)$ is a local minimum.

For g, we have $\nabla g = (2x, -2y)^{\mathsf{T}}$, so the only critical point is $(x,y) = (0,0)$.

Its Hessian is $H_f = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}$, with eigenvalues $\lambda = 2, -2$. The mixed signs give a saddle point.

Multivariable chain rule

Recall that for single-variable functions, the chain rule states that if $u = f(x)$ and $y = g(u) = g(f(x)) = (g \circ f)(x)$ then

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx} = g'(f(x))f'(x).$$

For multivariable functions, we also have the **multivariable chain rule**, which states that the Jacobian matrix of a composite function is equal to the product of the Jacobian matrices. Specifically, let $f : \mathbb{R}^d \to \mathbb{R}^m$ and $g : \mathbb{R}^m \to \mathbb{R}^k$ be differentiable. Then $g \circ f$ is differentiable and we have

$$\boxed{J_{g\circ f}(x) = J_g(f(x))J_f(x)}$$

In particular, if $z = f(x(t), y(t))$ and $\mathbf{r}(t) = (x(t), y(t))$, then

$$\boxed{\frac{dz}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt} = \nabla f \cdot \mathbf{r}'(t)}$$

Multivariable chain rule

Example: Let $f(x, y) = x^2 y$ and $\mathbf{r}(t) = (t, t^2)$. Then

$$\nabla f = (2xy,\ x^2), \qquad \mathbf{r}' = (1, 2t),$$

so

$$\frac{d}{dt} f(\mathbf{r}(t)) = 2xy(1) + x^2(2t) = (2t(t^2)(1) + (t^2)(2t) = 4t^3.$$

Example: If $f = f(x, y)$ and $u(r, \theta) = f(r\cos\theta,\ r\sin\theta)$, we have

$$u_r = f_x \cos\theta + f_y \sin\theta$$

and

$$u_\theta = -f_x r\sin\theta + f_y r\cos\theta.$$

Multivariable chain rule and linear approximation

We can apply the multivariable chain rule to derive the linear approximation:

$$f(x,y) \approx f(x_0,y_0) + \frac{\partial f}{\partial x}(x_0,y_0)(x-x_0) + \frac{\partial f}{\partial y}(x_0,y_0)(y-y_0)$$

1. Recall Single-Variable Linear Approximation

For a single-variable function $g(t)$ near t_0, the tangent line approximation is:

$$g(t_0+\Delta t) \approx g(t_0) + g'(t_0)\Delta t$$

2. Define a Single-Variable Slice

To approximate $f(x,y)$ at a nearby point $(x_0+\Delta x, y_0+\Delta y)$, set up a function of a single variable t moving along a straight path from (x_0,y_0) to $(x_0+\Delta x, y_0+\Delta y)$:

$$g(t) = f(x_0 + t\Delta x,\ y_0 + t\Delta y)$$

Notice that $g(0) = f(x_0,y_0)$ and $g(1) = f(x_0+\Delta x, y_0+\Delta y)$.

Multivariable chain rule and linear approximation

(Continued)

3. Apply the Chain Rule for Multivariable Functions

Find the derivative $g'(t)$ using the multivariable chain rule:

$$g'(t) = \frac{\partial f}{\partial x} \cdot \frac{d(x_0 + t\Delta x)}{dt} + \frac{\partial f}{\partial y} \cdot \frac{d(y_0 + t\Delta y)}{dt}$$

$$g'(t) = \frac{\partial f}{\partial x}(x_0 + t\Delta x, y_0 + t\Delta y)\Delta x + \frac{\partial f}{\partial y}(x_0 + t\Delta x, y_0 + t\Delta y)\Delta y$$

4. Evaluate at $t = 0$

At $t = 0$, the point $(x_0 + t\Delta x, y_0 + t\Delta y)$ becomes (x_0, y_0):

$$g'(0) = \frac{\partial f}{\partial x}(x_0, y_0)\Delta x + \frac{\partial f}{\partial y}(x_0, y_0)\Delta y$$

Multivariable chain rule and linear approximation

(Continued)

5. Substitute into the Single-Variable Approximation

Using $g(1) \approx g(0) + g'(0)(1-0)$:

$$f(x_0 + \Delta x, y_0 + \Delta y) \approx f(x_0, y_0) + \left[\frac{\partial f}{\partial x}(x_0, y_0)\Delta x + \frac{\partial f}{\partial y}(x_0, y_0)\Delta y\right]$$

6. Rewrite with x and y

Finally, substituting $\Delta x = x - x_0$ and $\Delta y = y - y_0$ yields the desired formula:

$$f(x, y) \approx f(x_0, y_0) + \frac{\partial f}{\partial x}(x_0, y_0)(x - x_0) + \frac{\partial f}{\partial y}(x_0, y_0)(y - y_0)$$

First-order Taylor approximation

More generally, if $f : \mathbb{R}^n \to \mathbb{R}$ is differentiable at $\mathbf{x}$, the **first-order Taylor approximation** is given by

$$\boxed{f(\mathbf{x} + \Delta\mathbf{x}) \approx f(\mathbf{x}) + \nabla f(\mathbf{x})^{\mathsf{T}} \Delta\mathbf{x}}$$

The approximation error is small relative to $\|\Delta x\|_2$ when $\Delta x \to \mathbf{0}$.

As a particular case, if $f = f(x, y)$ is a function of two variables, we have

$$f(x, y) \approx f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

Second-order Taylor approximation

If $f : \mathbb{R}^n \to \mathbb{R}$ is twice differentiable at $\mathbf{x}$, the **second-order Taylor approximation** is given by

$$\boxed{f(\mathbf{x} + \Delta\mathbf{x}) \approx f(\mathbf{x}) + \nabla f(\mathbf{x})^\mathsf{T} \Delta\mathbf{x} + \frac{1}{2} \Delta\mathbf{x}^\mathsf{T} H_f(\mathbf{x}) \Delta\mathbf{x}}$$

As a particular case, if $f = f(x, y)$ is a function of two variables, we have

$$\begin{aligned} f(x, y) \approx\ & f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) \\ & + \frac{1}{2} \left(f_{xx}(x_0, y_0)(x - x_0)^2 + 2 f_{xy}(x - x_0)(y - y_0) + f_{yy}(x_0, y_0)(y - y_0)^2 \right) \end{aligned}$$

Taylor approximation

More generally, we can have the Taylor series for multivariable functions

$$T(\mathbf{x}) = \sum_{n_1=0}^{\infty} \cdots \sum_{n_d=0}^{\infty} \frac{\left(\frac{\partial^{n_1+\cdots+n_d} f}{\partial x_1^{n_1} \cdots \partial x_d^{n_d}}\right)(\mathbf{a})}{n_1! \cdots n_d!} (x_1 - a_1)^{n_1} \cdots (x_d - a_d)^{n_d}$$
$$= \sum_{|\alpha| \geq 0} \frac{\mathrm{D}^\alpha f(\mathbf{a})}{\alpha!} (\mathbf{x} - \mathbf{a})^\alpha$$

and the k-th order Taylor approximation

$$\boxed{f(\boldsymbol{x}) \approx \sum_{|\alpha| \leq k} \frac{D^\alpha f(\boldsymbol{a})}{\alpha!} (\boldsymbol{x} - \boldsymbol{a})^\alpha}$$

where

$$D^\alpha f = \frac{\partial^{\alpha_1 + \ldots + \alpha_n} f}{\partial x_1^{\alpha_1} \cdots \partial x_n^{\alpha_n}}.$$

Taylor approximation

Example:
Let $f(x, y) = x^2 + xy + 2y^2$. To find the first and second-order Taylor approximation at $(1, 1)$, we have

$$f(1,1) = 4, \quad \nabla f(1,1) = (3,5)^\mathsf{T}, \quad H_f(1,1) = \begin{bmatrix} 2 & 1 \\ 1 & 4 \end{bmatrix},$$

and hence the first-order Taylor approximation is

$$f((1,1) + \Delta\mathbf{x}) \approx 4 + (3,5)^\mathsf{T}\Delta\mathbf{x}$$

and the second-order Taylor approximation is

$$f((1,1) + \Delta\mathbf{x}) \approx 4 + (3,5)^\mathsf{T}\Delta\mathbf{x} + \frac{1}{2}\Delta\mathbf{x}^\mathsf{T}\begin{bmatrix} 2 & 1 \\ 1 & 4 \end{bmatrix}\Delta\mathbf{x}.$$

Taylor approximation

(Continued)
Now consider $\Delta \mathbf{x} = (0.1, -0.2)^{\mathsf{T}}$. The first-order approximation gives

$$f(1.1, 0.8) \approx 4 + 3(0.1) + 5(-0.2) = 4 - 0.7 = 3.3$$

and second-order Taylor approximation gives

$$f(1.1, 0.8) \approx 4 - 0.7 + \tfrac{1}{2} \begin{bmatrix} 0.1 & -0.2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} 0.1 \\ -0.2 \end{bmatrix} = 4 - 0.7 + 0.07 = 3.37.$$

We can further calculate the actual value of $f(1.1, 0.8)$:

$$f(1.1, 0.8) = (1.1)^2 + (1.1)(0.8) + 2(0.8)^2 = 1.21 + 0.88 + 1.28 = 3.37.$$

The match is exact because f itself is a quadratic function.

Example

Find the linear and quadratic approximations of $f(x, y) = \dfrac{2}{xy}$ near $(1, 2)$.

Example

Find the linear and quadratic approximations of $f(x,y) = \dfrac{2}{xy}$ near $(1,2)$.

Answer: Note that $f(1,2) = 1$. We also compute derivatives at $(1,2)$:

$$f_x(x,y) = -\frac{2}{x^2y} \Rightarrow f_x(1,2) = -\frac{2}{1^2 \cdot 2} = -1,$$

$$f_y(x,y) = -\frac{2}{xy^2} \Rightarrow f_y(1,2) = -\frac{2}{1 \cdot 2^2} = -\frac{1}{2},$$

$$f_{xx}(1,2) = 2, \qquad f_{xy}(1,2) = \frac{1}{2}, \qquad f_{yy}(1,2) = \frac{1}{2},$$

and hence the linear approximation of $f(x,y)$ near $(1,2)$ is

$$f(x,y) \approx f(1,2) + f_x(1,2)(x-1) + f_y(1,2)(y-2) = 1 - (x-1) - \frac{1}{2}(y-2) = 3 - x - \frac{1}{2}y$$

Example

Find the linear and quadratic approximations of $f(x,y) = \dfrac{2}{xy}$ near $(1,2)$.

Answer: (Continued)

The quadratic approximation of $f(x,y)$ near $(1,2)$ is

$$\begin{aligned} f(x,y) &\approx f(1,2) + f_x(1,2)(x-1) + f_y(1,2)(y-2) \\ &\quad + \frac{1}{2}f_{xx}(1,2)(x-1)^2 + f_{xy}(1,2)(x-1)(y-2) + \frac{1}{2}f_{yy}(1,2)(y-2)^2 \\ &= 1 - (x-1) - \frac{1}{2}(y-2) + (x-1)^2 + \frac{1}{2}(x-1)(y-2) + \frac{1}{4}(y-2)^2 \\ &= 6 - 4x - 2y + x^2 + \frac{1}{2}xy + \frac{1}{4}y^2 \end{aligned}$$

Computation graphs

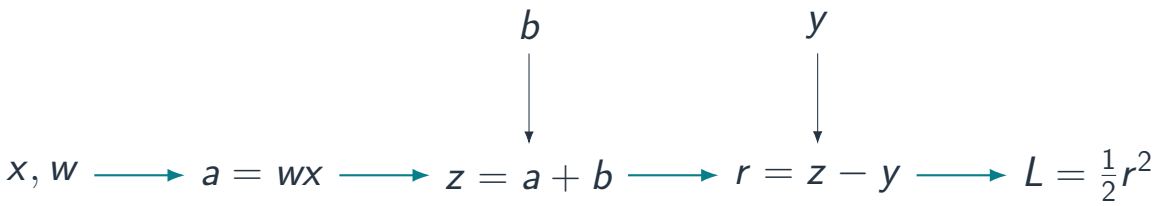
A **computation graph** is a directed network of nodes and edges used to break down and execute mathematical expressions step-by-step. In particular,

- The nodes represent values (like inputs, constants, or tensors) or operations (like addition, multiplication, or activation functions).
- The directed edges represent the directed flow of data, showing which values feed into which operations.

Example: Consider a one-observation squared loss:

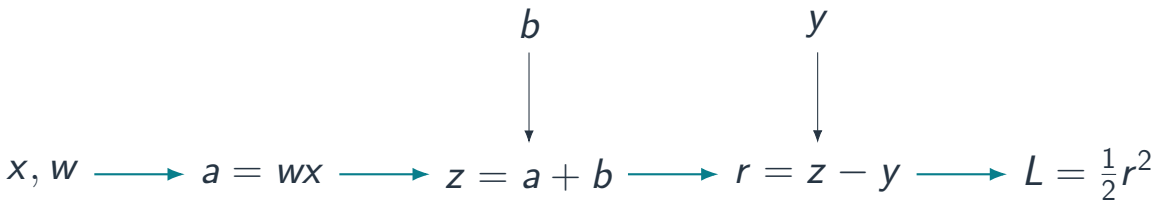
$$a = wx, \qquad z = a + b, \qquad r = z - y, \qquad L = \tfrac{1}{2}r^2.$$

We have



Computation graphs and automatic differentiation

Automatic differentiation (AD) is a set of techniques used by computer programs to evaluate the exact derivative of a function by breaking complex calculations down into basic elementary operations and applying the chain rule



Specifically, we have:

- The forward pass: We can compute the function values and derivative information as data moves forward through the graph from inputs to the final output through the operations.
- The backward pass: We can also run the function forward to record a computational graph, then sweep backward to calculate gradients by applying the chain rule.

Computation graphs and automatic differentiation

Example: Take $x = 2$, $w = 3$, $b = 1$, $y = 5$. We have the forward pass:

Node	Rule	Substitution	Value
a	wx	$3(2)$	6
z	$a + b$	$6 + 1$	7
r	$z - y$	$7 - 5$	2
L	$r^2/2$	$2^2/2$	2

These intermediate values will be reused in the backward pass.

Computation graphs and automatic differentiation

(Continued)

We can further get the local derivatives:

Operation	Local derivatives
$a = wx$	$\partial a/\partial w = x$, $\partial a/\partial x = w$
$z = a + b$	$\partial z/\partial a = 1$, $\partial z/\partial b = 1$
$r = z - y$	$\partial r/\partial z = 1$, $\partial r/\partial y = -1$
$L = r^2/2$	$\partial L/\partial r = r$

An operation's local derivative is different from its total influence on the final loss.

Computation graphs and automatic differentiation

More specifically, the **adjoint** of a scalar node in a computation graph is the partial derivative of the final scalar output L with respect to that node's value v, denoted as

$$\bar{v} = \frac{\partial L}{\partial v}$$

It measures how much a tiny change in that specific node affects the final output

Adjoints are computed during the backward pass (reverse-mode automatic differentiation or backpropagation) flowing from the output back to the inputs.

Computation graphs and automatic differentiation

Reverse mode automatic differentiation (back-propagation) procedure:

- **Initialization**: The output node L starts with an adjoint of

$$\bar{L} = \frac{\partial L}{\partial L} = 1$$

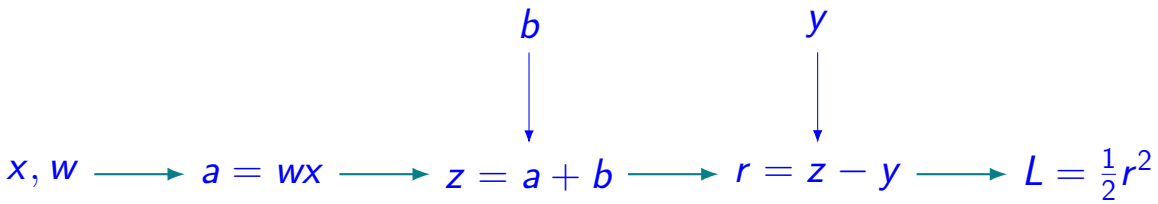
- **Chain Rule Accumulation**: For any node v_i, its adjoint $\bar{v}_i$ is the sum of the adjoints of all its child nodes (successors) v_j, weighted by the partial derivative of each child with respect to that node:

$$\bar{v}_i = \sum_{v_j \in \text{children}(v_i)} \bar{v}_j \frac{\partial v_j}{\partial v_i} = \sum_{v_j : v_i \to v_j} \bar{v}_j \frac{\partial v_j}{\partial v_i}.$$

- **Traversal**: The graph is visited in reverse topological order so that every child's adjoint is fully known before computing the parent's adjoint.

Computation graphs and automatic differentiation

Example: Based on the following computation graph, evaluate the adjoints at $x = 2$, $w = 3$, $b = 1$, $y = 5$.



First, we have

$$\bar{L} = \frac{\partial L}{\partial L} = 1$$

Since $L = r^2/2$,

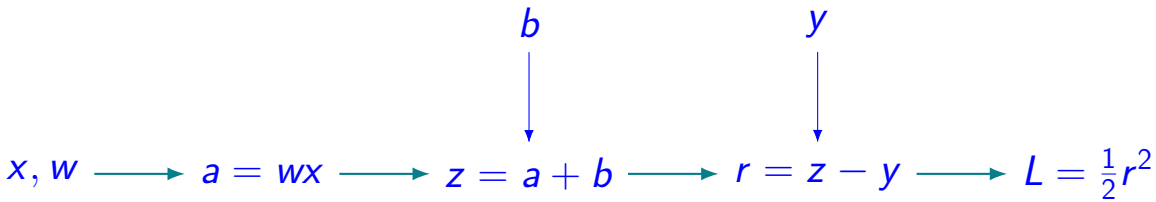
$$\bar{r} = \bar{L}\frac{\partial L}{\partial r} = 1 \cdot r = 2.$$

Since $r = z - y$,

$$\bar{z} = \bar{r}\frac{\partial r}{\partial z} = 2, \qquad \bar{y} = \bar{r}\frac{\partial r}{\partial y} = -2.$$

Computation graphs and automatic differentiation

(Continued)



From $z = a + b$,

$$\bar{a} = 2, \qquad \bar{b} = 2.$$

From $a = wx$,

$$\bar{w} = \bar{a}x = 2(2) = 4, \qquad \bar{x} = \bar{a}w = 2(3) = 6.$$

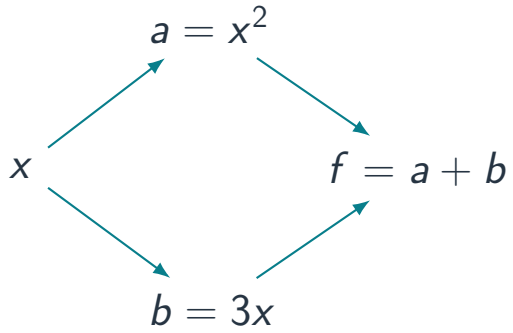
We can verify the result by direct differentiation. We can first express $L = \frac{1}{2}(wx + b - y)^2$, and direct differentiation gives

$$\frac{\partial L}{\partial w} = (wx + b - y)x,$$

At the chosen values, we have $\frac{\partial L}{\partial w} = ((3)(2) + 1 - 5)(2) = 4$, matching the graph calculation.

Branching creates multiple contributions

In some cases we will encounter branching:



Here $f = x^2 + 3x$ depends on x through two paths. One path contributes $2x$; the other contributes 3. Therefore $df/dx = 2x + 3$, not either contribution alone.

In automatic differentiation, we will also take care of that.

Branching creates multiple contributions

Example:
Consider the scalar map

$$f(x) \;=\; \sin(x^2)\,\exp(x^2) \;+\; \left(x^2+2\right).$$

Introduce the intermediate quantities

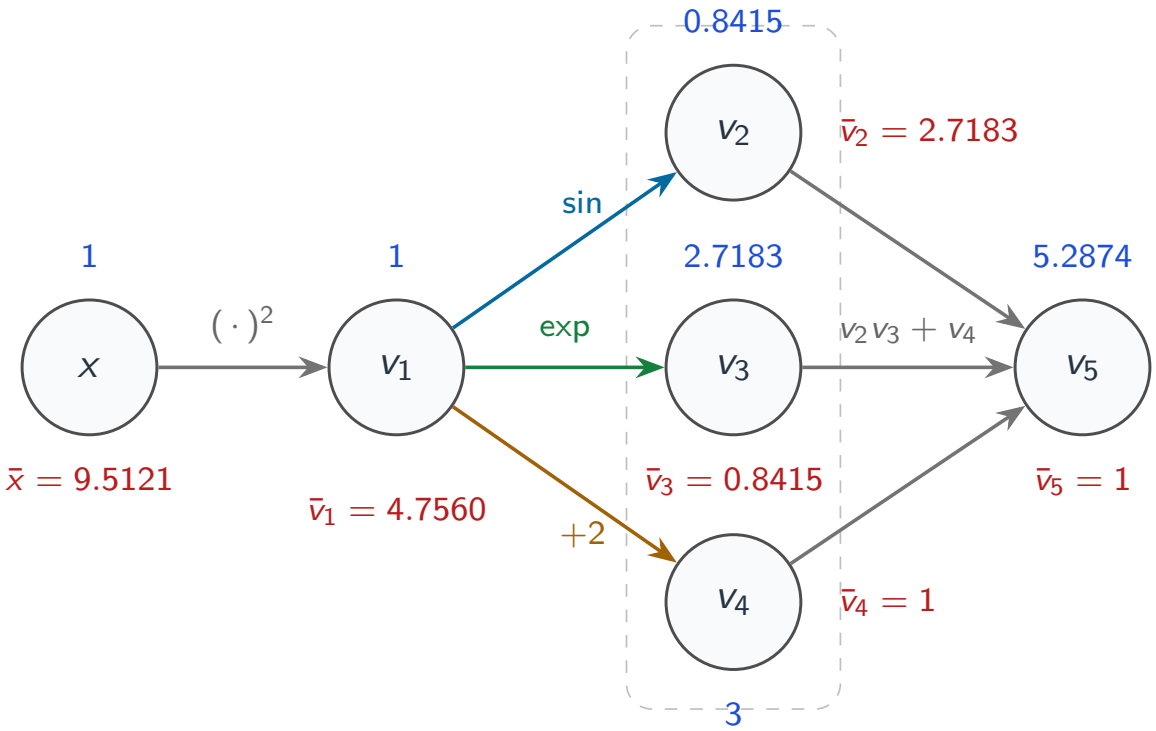
$$v_1 = x^2, \qquad v_2 = \sin(v_1), \qquad v_3 = \exp(v_1), \qquad v_4 = v_1 + 2, \qquad v_5 = v_2\,v_3 + v_4,$$

so that $f = v_5$. Our goal is to perform reverse-mode automatic differentiation to get $\bar{x}$ at $x = 1$.

Note that the node v_1 has **three outgoing branches**; those three local Jacobians must be accumulated when the adjoint $\bar{v}_1$ is formed.

We will construct the computation graph, perform forward sweep (primal evaluation), and then reverse sweep (adjoint propagation).

Branching creates multiple contributions



blue = primal values at $x = 1$ red = adjoints $\bar{v} = \partial f / \partial v$

Branching creates multiple contributions

(Continued)

Forward sweep (primal evaluation):

Compute all function values starting from the input $x = 1$:

$$
\begin{aligned}
v_1 &= x^2 = 1, \\
v_2 &= \sin(v_1) = \sin 1 \approx 0.8415, \\
v_3 &= \exp(v_1) = e \approx 2.7183, \\
v_4 &= v_1 + 2 = 3, \\
v_5 &= v_2 v_3 + v_4 \approx 5.2874 = f(1).
\end{aligned}
$$

The local partial derivatives that will be needed on the way back are stored (or recomputed) together with these primal values:

$$
\frac{\partial v_1}{\partial x} = 2x, \quad \frac{\partial v_2}{\partial v_1} = \cos v_1, \quad \frac{\partial v_3}{\partial v_1} = \exp v_1, \quad \frac{\partial v_4}{\partial v_1} = 1, \quad \frac{\partial v_5}{\partial v_2} = v_3, \quad \frac{\partial v_5}{\partial v_3} = v_2, \quad \frac{\partial v_5}{\partial v_4} = 1.
$$

Branching creates multiple contributions

(Continued)

Reverse sweep (adjoint propagation)

Initialize the output adjoint $\bar{v}_5 = 1$, and then walk the graph backwards.

For each node v, the adjoint is

$$\bar{v} = \sum_{w:\, v \to w} \bar{w} \frac{\partial w}{\partial v}.$$

From v_5 into the three branches:

$$\bar{v}_2 = \bar{v}_5 \frac{\partial v_5}{\partial v_2} = 1 \cdot v_3 \approx 2.7183,$$

$$\bar{v}_3 = \bar{v}_5 \frac{\partial v_5}{\partial v_3} = 1 \cdot v_2 \approx 0.8415,$$

$$\bar{v}_4 = \bar{v}_5 \frac{\partial v_5}{\partial v_4} = 1 \cdot 1 = 1.$$

Branching creates multiple contributions

(Continued)

Accumulation at node v_1:

This is the distinctive step: three incoming adjoint contributions are added.

$$\begin{aligned}\bar{v}_1 &= \bar{v}_2 \cos v_1 + \bar{v}_3 \exp v_1 + \bar{v}_4 \cdot 1 \\ &= 2.7183 \cdot \cos 1 + 0.8415 \cdot e + 1 \;\approx\; 4.7560.\end{aligned}$$

Back to the input:

$$\bar{x} = \bar{v}_1 \frac{\partial v_1}{\partial x} = 4.7560 \cdot 2x = 4.7560 \cdot 2 \approx 9.5121.$$

Hence

$$f'(1) \;=\; \bar{x} \;\approx\; 9.5121.$$

Branching creates multiple contributions

(Continued)
We can cross check the result against the closed-form derivative. Differentiating the formula directly,

$$f'(x) = 2x\Big(\mathrm{e}^{x^2}\big(\sin(x^2) + \cos(x^2)\big) + 1\Big).$$

Substituting $x = 1$ recovers exactly the same value $2\Big(e\big(\sin 1 + \cos 1\big) + 1\Big) \approx 9.5121$.

The three-branch accumulation at v_1 is therefore the chain rule, implemented by summing adjoints rather than by building a symbolic expression.

Next time

Lectures 5–8 will be taught by Dr. Simon Cheung

- Probability Foundations
- Information Theory
- Statistical Inference
- Bayesian Statistics and Bayesian Networks